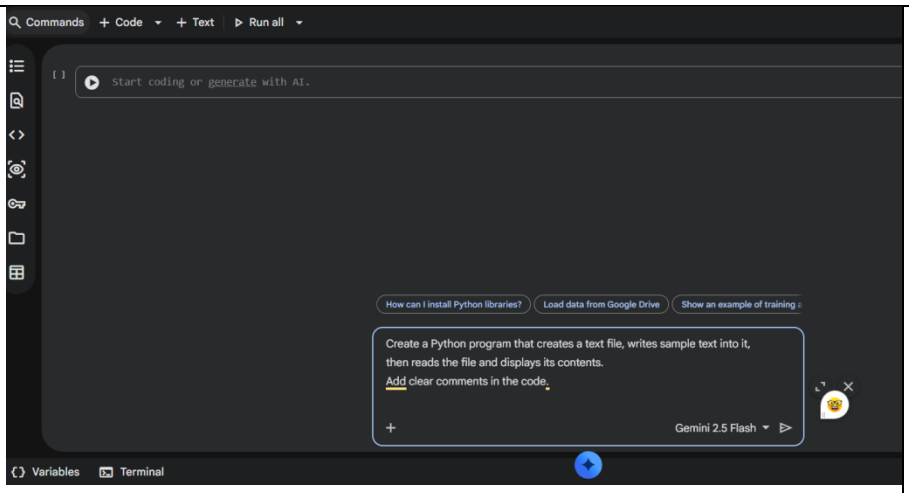
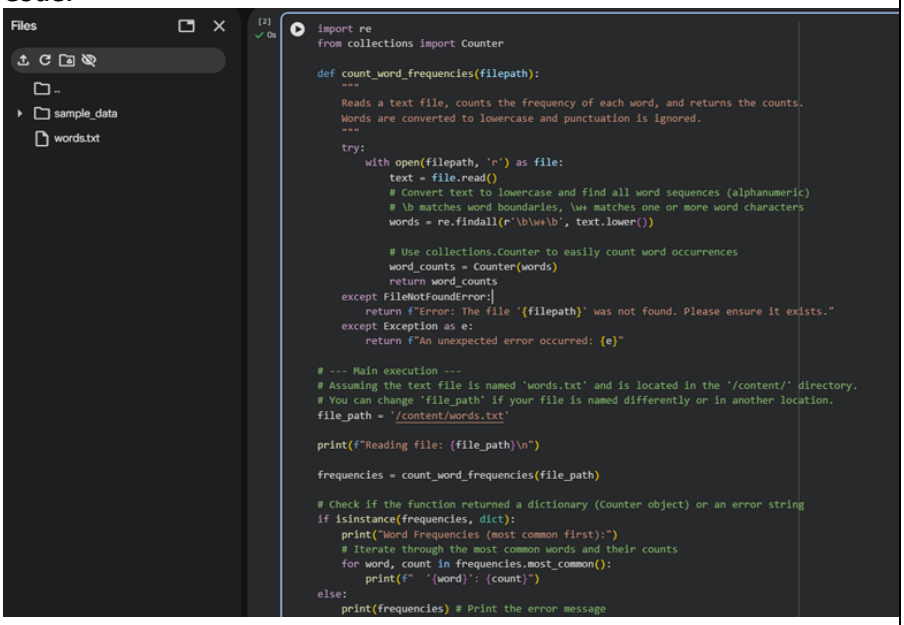
A. Supreeth Reddy
2303A51392
Batch 06
AIAC Lab 2.3

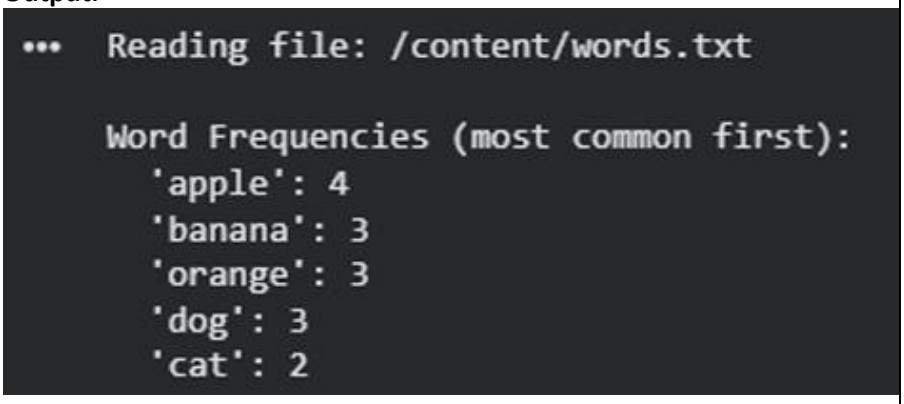
Q.No.	Question	
1	Lab 2: Exploring Additional AI Coding Tools beyond Copilot – Gemini (Colab) and Cursor AI **Lab Objectives:** ❖ To explore and evaluate the functionality of Google Gemini for AI-assisted coding within Google Colab. ❖ To understand and use Cursor AI for code generation, explanation, and refactoring. ❖ To compare outputs and usability between Gemini, GitHub Copilot, and Cursor AI. ❖ To perform code optimization and documentation using AI tools. **Lab Outcomes (LOs):** After completing this lab, students will be able to: ❖ Generate Python code using Google Gemini in Google Colab. ❖ Analyze the effectiveness of code explanations and suggestions by Gemini. ❖ Set up and use Cursor AI for AI-powered coding assistance. ❖ Evaluate and refactor code using Cursor AI features. ❖ Compare AI tool behaviour and code quality across different platforms. **Task 1: Word Frequency from Text File** ❖ **Scenario:** You are analyzing log files for keyword frequency. ❖ **Task:** Use Gemini to generate Python code that reads a text file and counts word frequency, then explains the code. ❖ **Prompt:** Create a Python program that creates a text file, writes sample text into it, then reads the file and displays its contents. Add clear comments in the code.	


Commands
+ Code
+ Text
Run all
Start coding or generate with AI.
How can I install Python libraries?
Load data from Google Drive
Show an example of training a
Create a Python program that creates a text file, writes sample text into it, then reads the file and displays its contents.
Add clear comments in the code.
Gemini 2.5 Flash
Variables
Terminal

Code:


Files
..
sample_data
words.txt
import re
from collections import Counter
def count_word_frequencies(filepath):
"""
Reads a text file, counts the frequency of each word, and returns the counts.
Words are converted to lowercase and punctuation is ignored.
"""
try:
with open(filepath, 'r') as file:
text = file.read()
# Convert text to lowercase and find all word sequences (alphanumeric)
# \b matches word boundaries, \w+ matches one or more word characters
words = re.findall(r'\b\w+\b', text.lower())
# Use collections.Counter to easily count word occurrences
word_counts = Counter(words)
return word_counts
except FileNotFoundError:
return f"Error: The file '{filepath}' was not found. Please ensure it exists."
except Exception as e:
return f"An unexpected error occurred: {e}"
# --- Main execution ---
# Assuming the text file is named 'words.txt' and is located in the '/content/' directory.
# You can change 'file_path' if your file is named differently or in another location.
file_path = '/content/words.txt'
print(f"Reading file: {file_path}\n")
frequencies = count_word_frequencies(file_path)
# Check if the function returned a dictionary (Counter object) or an error string
if isinstance(frequencies, dict):
print("Word Frequencies (most common first):")
# Iterate through the most common words and their counts
for word, count in frequencies.most_common():
print(f" '{word}': {count}")
else:
print(frequencies) # Print the error message

Output:

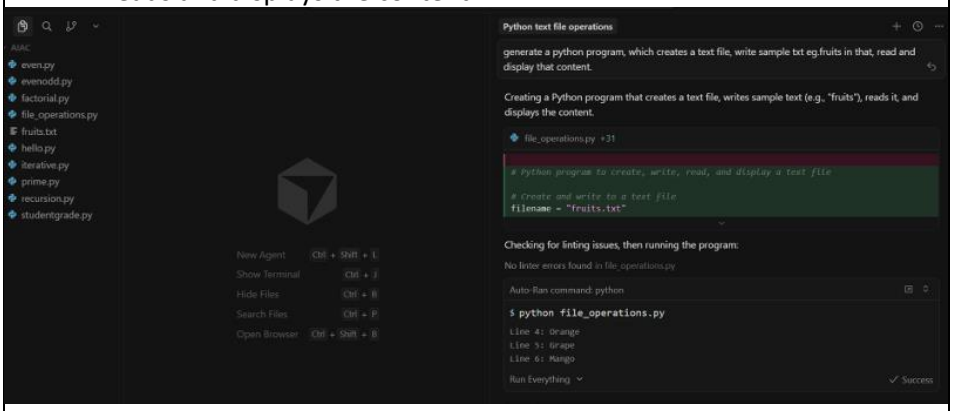

Reading file: /content/words.txt
Word Frequencies (most common first):
'apple': 4
'banana': 3
'orange': 3
'dog': 3
'cat': 2

Explanation:
This program reads a text file and goes through its contents word by word.
As it reads, it keeps track of each word and remembers how many times it appears.
If the same word shows up again, the program simply increases its count.
Once the entire file is processed, it prints out each word along with how often it was used.
This makes it easy to understand which words appear most frequently in the text.

Task 2: File Operations Using Cursor AI

- **Scenario:**
  You are automating basic file operations.

- **Task:**
  Use Cursor AI to generate a program that:
  - Creates a text file
  - Writes sample text
  - Reads and displays the content



Code:

file_operations.py > ...

```python
# Python program to create, write, read, and display a text file
# Create and write to a text file
filename = "fruits.txt"

# Write sample text to the file
with open(filename, 'w') as file:
    file.write("fruits\n")
    file.write("Apple\n")
    file.write("Banana\n")
    file.write("Orange\n")
    file.write("Grape\n")
    file.write("Mango\n")

print(f"File '{filename}' has been created and written to.\n")

# Read and display the content
print("Reading and displaying the content:")
print("-" * 40)
with open(filename, 'r') as file:
    content = file.read()
    print(content)
print("-" * 40)

# Alternative: Read line by line
print("\nReading line by line:")
print("-" * 40)
with open(filename, 'r') as file:
    for line_num, line in enumerate[str](file, 1):
        print(f"Line {line_num}: {line.strip()}")
```

Output:

```
Reading and displaying the content:
----------------------------------------
fruits
Apple
Banana
Orange
Grape
Mango

----------------------------------------

Reading line by line:
----------------------------------------
Line 1: fruits
Line 2: Apple
Line 3: Banana
Line 4: Orange
Line 5: Grape
Line 6: Mango
```

Explanation:
This program shows how basic file handling works in Python using a cursor.
It starts by creating a text file and writing some sample text into it.
After that, the file is opened again in read mode to display its contents on the screen.
Through this process, the program demonstrates how easily Python can create, write to, and read from files.
These file operations are commonly used in tasks like automation and simple data storage.

Task 3: CSV Data Analysis

- **Scenario:**
  You are processing structured data from a CSV file.

- **Task:**
  Use Gemini in Colab to read a CSV file and calculate mean, min, and max.

- **Prompt:**

Generate Python code to read a CSV file and calculate the mean, minimum, and maximum values of a numeric column.

Gemini 2.5 Flash

Code:

```python
import pandas as pd
import io

# Uncomment the lines below and replace 'your_file.csv' with your CSV file path
try:
    df = pd.read_csv('scores.csv')
except FileNotFoundError:
    print("Error: CSV file not found. Please check the path.")
    exit()

# Specify the numeric column you want to analyze
column_name = 'score'

# Check if the column exists in the DataFrame
if column_name in df.columns:
    # Ensure the column is numeric (e.g., float or int)
    # pd.to_numeric will convert values to numeric, coercing errors to NaN
    numeric_column = pd.to_numeric(df[column_name], errors='coerce')

    # Drop rows where the numeric conversion resulted in NaN (non-numeric values)
    numeric_column = numeric_column.dropna()

    if not numeric_column.empty:
        # Calculate mean, minimum, and maximum
        mean_value = numeric_column.mean()
        min_value = numeric_column.min()
        max_value = numeric_column.max()

        print(f"Statistics for column '{column_name}':")
        print(f"  Mean: {mean_value:.2f}")
        print(f"  Minimum: {min_value:.2f}")
        print(f"  Maximum: {max_value:.2f}")
    else:
        print(f"Column '{column_name}' contains no valid numeric data after cleaning.")
else:
    print(f"Error: Column '{column_name}' not found in the CSV file. Available columns: {df.columns.tolist()}")
```

❖ **Expected Output:**

```
Statistics for column 'score':
  Mean: 82.00
  Minimum: 67.00
  Maximum: 92.00
```

Explanation:

This program reads data from a CSV file using Python.
It focuses on a column of numerical values and calculates useful statistics such as the average (mean), the smallest value, and the largest value.
By doing this, the program shows how CSV files can be analyzed efficiently.
Such analysis is commonly used in data processing and analytics applications.

Task 4: Sorting Lists – Manual vs Built-in

- **Scenario:**
  You are reviewing algorithm choices for efficiency.

- **Task:**
  Use **Gemini** to generate:
  - Bubble sort
  - Python's built-in sort()
  - Compare both implementations.

Prompt:

K Generate Python code for Bubble Sort

Empty cell

Here's the Python code for implementing the Bubble Sort algorithm. After the code, I'll explain how it works in simple terms.

Sources

1. github.com/rbernalop/codemmunity

K Generate Python code for built-in sort

Certainly! Python provides excellent built-in ways to sort lists. Here are examples using both the `list.sort()` method and the `sorted()` function, along with an explanation of their differences.

Code:

Manual Bubble Sort Code:

```python
def bubble_sort(arr):
    n = len(arr)
    # Traverse through all array elements
    for i in range(n):
        # Last i elements are already in place
        for j in range(0, n - i - 1):
            # Traverse the array from 0 to n-i-1
            # Swap if the element found is greater than the next element
            if arr[j] > arr[j + 1]:
                arr[j], arr[j + 1] = arr[j + 1], arr[j] # Swap elements
    return arr

# --- Example Usage ---
my_list = [64, 34, 25, 12, 22, 11, 90]
print("Original list:", my_list)

sorted_list = bubble_sort(my_list)
print("Sorted list using Bubble Sort:", sorted_list)

my_list_2 = [5, 1, 4, 2, 8]
print("\nOriginal list 2:", my_list_2)

sorted_list_2 = bubble_sort(my_list_2)
print("Sorted list 2 using Bubble Sort:", sorted_list_2)
```

Output:

```
Original list: [64, 34, 25, 12, 22, 11, 90]
Sorted list using Bubble Sort: [11, 12, 22, 25, 34, 64, 90]

Original list 2: [5, 1, 4, 2, 8]
Sorted list 2 using Bubble Sort: [1, 2, 4, 5, 8]
```

Sorting using sort() function:

```python
my_list_1 = [64, 34, 25, 12, 22, 11, 90]
print("Original list (list.sort()):", my_list_1)

my_list_1.sort()
print("Sorted list (list.sort()):", my_list_1)

# You can also sort in descending order
my_list_2 = ['banana', 'apple', 'cherry', 'date']
print("\nOriginal list (descending):", my_list_2)

my_list_2.sort(reverse=True)
print("Sorted list (descending):", my_list_2)
```

Output:

```
Original list (list.sort()): [64, 34, 25, 12, 22, 11, 90]
Sorted list (list.sort()): [11, 12, 22, 25, 34, 64, 90]

Original list (descending): ['banana', 'apple', 'cherry', 'date']
Sorted list (descending): ['date', 'cherry', 'banana', 'apple']
```

	Comparison: The sort() function is highly efficient and best suited for real-world programs because it is fast, reliable, and easy to use. Manual Bubble Sort, on the other hand, is mainly used for educational purposes to understand how sorting works internally, but it is not practical for large datasets due to its poor performance.	